

Information Bottlenecks and Feature Emergence

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Abstract

This research explores the trade-offs between dimensionality reduction and feature preservation within the context of deterministic Autoencoders (AEs). Utilizing the Fashion-MNIST dataset, we investigate how varying latent space capacities (4, 32, and 128 dimensions) influence the model’s ability to transition from pixel-level reconstruction to semantic representation. We demonstrate that while overcomplete models (Latent 128) minimize Mean Squared Error (MSE), they fail to learn a generalized manifold, whereas undercomplete models (Latent 32) act as optimal denoisers. Furthermore, we evaluate the "Upstream" utility of these representations through a "Downstream" linear probing task, revealing the geometric separability of the learned latent spaces.

Theoretical Framework: The Manifold Hypothesis

The fundamental premise of this study is the **Manifold Hypothesis**, which posits that high-dimensional data (like 784-dimensional images) actually lies on a much lower-dimensional, non-linear manifold.

An Autoencoder (AE) is a neural network designed to learn this manifold through an *Information Bottleneck*. The model consists of two functions:

1. **The Encoder (f_ϕ):** Maps the input $x \in \mathbb{R}^{784}$ to a latent vector $z \in \mathbb{R}^d$.
2. **The Decoder (g_θ):** Maps the latent vector z back to the reconstruction $\hat{x} \in \mathbb{R}^{784}$.

The objective is to minimize the reconstruction loss $\mathcal{L}$:

$$\min_{\theta, \phi} \mathbb{E}_{x \sim P_{data}} [\|x - g_\theta(f_\phi(x))\|^2] \quad (1)$$

Task-by-Task Experimental Methodology

Task 1: Statistical Variance in Fashion-MNIST

Before modeling, the data distribution was inspected. Fashion-MNIST is notably more complex than MNIST because classes share global silhouettes (e.g., Coats, Pullovers, and Shirts). **Intuition:** Successful representation learning requires the model to prioritize "Discriminative Features" (buttons, sleeve length) over "Non-Discriminative Features" (average brightness).

Task 2 & 3: Architectural Capacity and Loss Analysis

We architected three MLP-based Autoencoders with latent dimensions $d \in \{4, 32, 128\}$.

- **Undercomplete (d=4):** Extreme compression (196:1 ratio). Forces the model to find only the most primitive global structures.

- **Balanced (d=32):** Optimized compression (24:1 ratio).
- **Overcomplete (d=128):** Low compression (6:1 ratio). Risks "Identity Mapping" where $f(x) \approx x$.

Results Observed from Notebook: The Latent 128 model achieved the lowest training loss (0.0119), yet the Latent 32 model produced reconstructions that were visually indistinguishable to the human eye, suggesting that much of the "extra" information captured by the 128-dim model is simply pixel-level noise.

The Geometry of Latent Space (Tasks 4 & 5)

Manifold Visualization (PCA)

By projecting z into 2D using Principal Component Analysis (PCA), we observe the topology of the learned brain. **Discovery:** In the 4-dim model, clusters for "Sandal" and "Sneaker" are physically overlapped. This is known as **Entanglement**. In the 32-dim model, these clusters move to distinct neighborhoods, proving the latent space has successfully "unfolded" the data.

Linear Interpolation: The Continuity Test

We test the manifold continuity by moving along a straight line in latent space:

$$z_{interp} = (1 - \alpha)z_1 + \alpha z_2, \quad \alpha \in [0, 1] \quad (2)$$

Intuition: If the transition between a "Dress" and a "Coat" looks like a realistic hybrid garment, the model has learned a continuous manifold. If it looks like a ghostly "fade-in/fade-out," the space is discontinuous and mathematically "sparse."

Stress Testing and Feature Utility (Tasks 6 & 7)

Noise Robustness: The Low-Pass Filter Effect

We subjected models to random and structured Gaussian noise. **Result:** The 4-dim model acts as an aggressive **Low-Pass Filter**. Because the bottleneck is too small to store "static," the model only reconstructs the clean silhouette. The 128-dim model, conversely, reconstructs the noise itself, failing the robustness test.

Task 7: The Downstream Utility Paradigm

Concept: Upstream vs. Downstream Tasks

The Autoencoder's reconstruction is the ****Upstream Task**** (General Education). The ****Downstream Task**** is Classification (Specialization). We use the Downstream task to measure how much "Utility" the Upstream task actually provided. If the Upstream task learned a good representation, a tiny classifier should easily solve the Downstream task.

The Gradient Detach Logic

To probe the representation, we train a linear layer $y = Wz + b$. **The Logic of `z.detach()`:** In PyTorch, gradients flow backward through the computational graph. If we do not call `.detach()`, the classifier's error would change the Encoder's weights. **Mathematical Reasoning:** We want to evaluate the *intrinsic* quality of the frozen latent space. Allowing the classifier to update the encoder would be "cheating," as the encoder would then optimize specifically for classification rather than demonstrating what it learned through reconstruction.

Final Analysis: Theoretical Synthesis (Task 8)

What information disappears first as the bottleneck tightens?

As we reduce latent size, high-frequency spatial information (logos, fine textures, sharp edges) disappears first. The network minimizes MSE by securing the low-frequency global silhouette, which accounts for the majority of the pixel-wise variance.

The Hyperplane Intuition: Why use a Linear Probe?

Geometric Separation:

Imagine the latent space as a 3D room filled with points of 10 different colors. A **Linear Classifier** is like a flat sheet of paper. If the colors are all mixed together, you cannot separate them with a flat sheet. If the Encoder has done its job, it has "pushed" all the blue points to one corner and all the red points to another. A Linear Probe measures how "flat" and "simple" the boundaries between these categories have become.

When does the latent space become meaningless?

The space becomes meaningless when it is **Sparse**. In deterministic AEs, the space between training points is "undefined." If you sample a random z coordinate, the Decoder produces shadowy, incoherent static. This proves that while the model knows the data points, it does not know the "laws" of the data distribution, a problem later solved by Variational Autoencoders (VAEs).

Conclusion

This study proves that **Compression is Intelligence**. The 32-dimensional model, while having 4x less capacity than the 128-dim model, developed a more robust and linearly separable representation. The overcomplete model demonstrated the dangers of memorization (overfitting), whereas the balanced undercomplete model successfully mapped the Fashion-MNIST data manifold.